1998–2022 Hong Kong (China) Mathematical Olympiad

Contents

Problems

Year 1998

1. $PQRS$ is a cyclic quadrilateral with $\angle PSR = 90°$. H, K are the feet of the perpendiculars from Q to PR, PS (suitably extended if necessary) respectively. Show that HK bisects QS.

2. The base of a pyramid is a convex polygon with 9 sides. Each of the diagonals of the base and each of the edges on the lateral surface of the pyramid is coloured either black or white. Both colours are used. (Note that the sides of the base are not coloured.) Prove that there are three segments coloured the same colour which form a triangle.

3. Let s, t be given nonzero integers, and let (x, y) be any ordered pair of integers. A move changes (x, y) to $(x + t, y - s)$. The pair (x, y) is 'good' if after some (maybe zero) number of moves it describes a pair of integers that are not relatively prime.

   (a) Determine if (s, t) is a good pair.

   (b) Show that for any s and t there is a pair (x, y) which is not good.

4. Let f be a function defined on the positive reals with the following properties:

   (1) $f(1) = 1$,

   (2) $f(x + 1) = xf(x)$, and

   (3) $f(x) = 10^{g(x)}$,

   where $g(x)$ is a function defined on the reals satisfying

   $$g(ty + (1 - t)z) \leqslant tg(y) + (1 - t)g(z)$$

   for all y and z and for $0 \leqslant t \leqslant 1$.

   (a) Prove that $t[g(n) - g(n - 1)] \leqslant g(n + t) - g(n) \leqslant t[g(n + 1) - g(n)]$ where n is an integer and $0 \leqslant t \leqslant 1$.

   (b) Prove that $\dfrac{4}{3} \leqslant f\left(\dfrac{1}{2}\right) \leqslant \dfrac{4}{3}\sqrt{2}$.

Year 1999

1. Determine all positive rational numbers $r \neq 1$ such that $r^{\frac{1}{r-1}}$ is rational.

2. Let I and O be the incentre and circumcentre of $\triangle ABC$, respectively. Assume $\triangle ABC$ is not equilateral (so $I \neq O$). Prove that $\angle AIO \leqslant 90^\circ$ if and only if $2BC \leqslant AB + CA$.

3. Students have taken a test paper in each of n ($n \geqslant 3$) subjects. It is known that for any subject exactly three students get the best score in the subject, and for any two subjects exactly one student gets the best score in every one of these two subjects. Determine the smallest n so that the above conditions imply that exactly one student gets the best score in every one of the n subjects.

4. Determine all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x + yf(x)) = f(x) + xf(y)$ for all $x, y \in \mathbb{R}$.

Year 2000

1. Let O be the circumcentre of $\triangle ABC$. Suppose $AB > AC > BC$. Let D be a point on the minor arc BC such that AD is not a diameter of the circumcircle of $\triangle ABC$. Let E and F be points on AD such that $AB \perp OE$ and $AC \perp OF$. Let P be the intersection of BE and CF. If $PB = PC + PO$, then prove that $\angle BAC = 30°$.

2. Let $a_1 = 1$, $a_{n+1} = \dfrac{a_n}{n} + \dfrac{n}{a_n}$ for $n = 1, 2, 3, \ldots$. Find the greatest integer less than or equal to a_{2000}. Be sure to prove your claim.

3. Find all prime numbers p and q such that $\dfrac{(7^p - 2^p)(7^q - 2^q)}{pq}$ is an integer.

4. A *lattice point* on the coordinate plane is a point with integer coordinates. Find all positive integer $n \geqslant 3$ such that there exists an n-sided polygon having lattice points as vertices and all sides have equal length.

Year 2001

1. A triangle ABC is given. A circle Γ passes through vertex A and is tangent to side BC at point P. The circle Γ intersects sides AB and AC at points M and N, respectively. Prove that (minor) arcs $\widehat{MP}$ and $\widehat{NP}$ are equal if and only if Γ is tangent to the circumcircle of $\triangle ABC$ at A.

2. Find all positive integers n such that the equation $x^3 + y^3 + z^3 = nx^2y^2z^2$ has positive integer solutions. Be sure to give a proof.

3. For each integer $k \geqslant 4$, prove that if $F(x)$ is a polynomial with integer coefficients satisfying the condition $0 \leqslant F(c) \leqslant k$ for every $c = 0, 1, \ldots, k+1$, then
$$F(0) = F(1) = \cdots = F(k+1).$$

4. There are 212 points inside or on a circle with radius 1. Prove that there are at least 2001 pairs of these points having distances at most 1.

Year 2002

1. Two circles intersect at points A and B. Through the point B a straight line is drawn, intersecting the first circle at K and the second circle at M. A line parallel to AM is tangent to the first circle at Q. The line AQ intersects the second circle again at R.

   (a) Prove that the tangent to the second circle at R is parallel to AK.

   (b) Prove that these two tangents are concurrent with KM.

2. Let $n \geqslant 3$ be an integer. In a conference there are n mathematicians. Every pair of mathematicians communicate in one of the n official languages of the conference. For any three different official languages, there exist three mathematicians who communicate with each other in these three languages. Determine all n for which this is possible. Justify your claim.

3. If $a \geqslant b \geqslant c \geqslant 0$ and $a+b+c=3$, then prove that $ab^2+bc^2+ca^2 \leqslant \dfrac{27}{8}$ and determine the equality case(s).

4. Let p be an odd prime such that $p \equiv 1 \pmod 4$. Evaluate with reasons, $\displaystyle\sum_{k=1}^{\frac{p-1}{2}} \left\{\frac{k^2}{p}\right\}$, where $\{x\} = x - [x]$, $[x]$ being the greatest integer not exceeding x.

Year 2003

1. Find the greatest real K such that for every positive u, v and w with $u^2 > 4vw$, the inequality
$$(u^2 - 4vw)^2 > K(2v^2 - uw)(2w^2 - uv)$$
holds. Justify your claim.

2. Let $ABCDEF$ be a regular hexagon of side length 1, and O be the centre of the hexagon. In addition to the sides of the hexagon, line segments are drawn from O to each vertex, making a total of twelve unit line segments. Find the number of paths of length 2003 along these line segments that start at O and terminate at O.

3. Let $ABCD$ be a cyclic quadrilateral. K, L, M, N are midpoints of sides AB, BC, CD and DA respectively. Prove that the orthocentres of triangles AKN, BKL, CLM, DMN are vertices of a parallelogram.

4. Find, with reasons, all integers a, b, and c such that
$$\frac{1}{2}(a+b)(b+c)(c+a) + (a+b+c)^3 = 1 - abc.$$

Year 2004

1. For $n \geqslant 2$, let $a_1, a_2, \ldots, a_n, a_{n+1}$ be positive and $a_2 - a_1 = a_3 - a_2 = \cdots = a_{n+1} - a_n \geqslant 0$. Prove that
$$\frac{1}{a_2^2} + \frac{1}{a_3^2} + \cdots + \frac{1}{a_n^2} \leqslant \frac{n-1}{2} \cdot \frac{a_1 a_n + a_2 a_{n+1}}{a_1 a_2 a_n a_{n+1}}.$$
Determine when equality holds.

2. In a school there are b teachers and c students. Suppose that

   (i) each teacher teaches exactly k students; and

   (ii) for each pair of distinct students, exactly h teachers teach both of them.

   Show that
$$\frac{b}{h} = \frac{c(c-1)}{k(k-1)}.$$

3. On the sides AB and AC of triangle ABC, there are points P and Q respectively such that $\angle APC = \angle AQB = 45°$. Let the perpendicular line to side AB through P intersects line BQ at S. Let the perpendicular line to side AC through Q intersects line CP at R. Let D be on side BC such that $AD \perp BC$. Prove that the lines PS, AD, QR meet at a common point and lines SR and BC are parallel.

4. Let $S = \{1, 2, \ldots, 100\}$. Determine the number of functions $f : S \to S$ satisfying the following conditions.

   (i) $f(1) = 1$;

   (ii) f is bijective (i.e. for every y in S, the equation $f(x) = y$ has exactly one solution);

   (iii) $f(n) = f(g(n))f(h(n))$ for every n in S.

   Here $g(n)$ and $h(n)$ denote the uniquely determined positive integers such that $g(n) \leqslant h(n)$, $g(n)h(n) = n$ and $h(n) - g(n)$ is as small as possible. (For instance, $g(80) = 8$, $h(80) = 10$ and $g(81) = h(81) = 9$.)

Year 2005

1. On a planet there are $3 \times 2005!$ aliens and 2005 languages. Each pair of aliens communicates with each other in exactly one language. Show that there are 3 aliens who communicate with each other in one common language.

2. Suppose there are $4n$ line segments of unit length inside a circle of radius n. Furthermore, a straight line L is given. Prove that there exists a straight line L' that is either parallel or perpendicular to L and that L' cuts at least two of the given line segments.

3. Let a, b, c, d be positive real numbers such that $a + b + c + d = 1$. Prove that

$$6(a^3 + b^3 + c^3 + d^3) \geqslant (a^2 + b^2 + c^2 + d^2) + \frac{1}{8}.$$

4. Show that there exist infinitely many squarefree positive integers n that divide $2005^n - 1$. (An integer is squarefree if it contains no factors of the form d^2, $d > 1$.)

Year 2006

1. Let M be a subset of $\{1, 2, \ldots, 2006\}$ with the following property: For any three elements x, y and z $(x < y < z)$ of M, $x + y$ does not divide z. Determine the largest possible size of M. Justify your claim.

2. For a positive integer k, let $f_1(k)$ be the square of the sum of the digits of k. (For example $f_1(123) = (1 + 2 + 3)^2 = 36$.) Let $f_{n+1}(k) = f_1(f_n(k))$. Determine the value of $f_{2007}(2^{2006})$. Justify your claim.

3. A convex quadrilateral $ABCD$ with $AC \neq BD$ is inscribed in a circle with centre O. Let E be the intersection of diagonals AC and BD. If P is a point inside $ABCD$ such that $\angle PAB + \angle PCB = \angle PBC + \angle PDC = 90°$, prove that O, P and E are collinear.

4. Let $a_1, a_2, a_3, \ldots$ be a sequence of positive numbers. If there exists a positive number M such that for every $n = 1, 2, 3, \ldots$,
$$a_1^2 + a_2^2 + \cdots + a_n^2 < M a_{n+1}^2,$$
then prove that there exists a positive number M' such that for every $n = 1, 2, 3, \ldots$,
$$a_1 + a_2 + \cdots + a_n < M' a_{n+1}.$$

Year 2007

1. Let D be a point on the side BC of triangle ABC such that $AB+BD = AC+CD$. The line segment AD cuts the incircle of triangle ABC at X and Y with X closer to A. Let E be the point of contact of the incircle of triangle ABC on the side BC. Show that

   (i) EY is perpendicular to AD,

   (ii) XD is $2IA'$, where I is the incentre of the triangle ABC and A' is the midpoint of BC.

2. Is there a polynomial f of degree 2007 with integer coefficients, such that $f(n), f(f(n)), f(f(f(n))), \ldots$ are pairwise relatively prime for every integer n? Justify your claim.

3. In a school there are 2007 male and 2007 female students. Each student joins not more than 100 clubs in the school. It is known that any two students of opposite genders have joined at least one common club. Show that there is a club with at least 11 male and 11 female members.

4. Determine if there exists a positive integer pair (m, n), such that

   (i) the greatest common divisor of m and n is 1, and $m \leqslant 2007$,

   (ii) for any $k = 1, 2, \ldots, 2007$, $\left[\dfrac{nk}{m}\right] = [\sqrt{2}k]$.

   (Here $[x]$ stands for the greatest integer less than or equal to x.)

Year 2008

1. Let $f(x) = c_m x^m + c_{m-1}x^{m-1} + \cdots + c_1 x + c_0$, where each c_i is a nonzero integer. Define a sequence $\{a_n\}$ by $a_1 = 0$ and $a_{n+1} = f(a_n)$ for all positive integers n.

   (a) Let i and j be positive integers with $i < j$. Show that $a_{j+1} - a_j$ is a multiple of $a_{i+1} - a_i$.

   (b) Show that $a_{2008} \neq 0$.

2. Let $n > 4$ be a positive integer such that n is composite (not a prime) and divides $\varphi(n)\sigma(n) + 1$, where $\varphi(n)$ is the Euler's totient function of n and $\sigma(n)$ is the sum of the positive divisors of n. Prove that n has at least three distinct prime factors.

3. The incircle of a scalene and acute $\triangle ABC$ touches BC, CA and AB at D, E and F respectively. H is a point on the segment EF such that $DH \perp EF$. Suppose $AH \perp BC$, prove that H is the orthocentre of $\triangle ABC$.

4. There are 2008 congruent circles on a plane such that no two are tangent to each other and each circle intersects at least three other circles. Let N be the total number of intersection points of these circles. Determine the smallest possible value of N.

Year 2009

1. Suppose $\{a_n\}$ is a sequence in which all the terms are integers, and a_2 is odd. For any natural number n, $n(a_{n+1} - a_n + 3) = a_{n+1} + a_n + 3$. Furthermore, a_{2009} is divisible by 2010. Find the smallest integer n, $n \geqslant 2$, such that a_n is divisible by 2010.

2. There are n points on the plane, no three of which are collinear. Each pair of points is joined by a red, yellow or green line. For any three points, the sides of the triangle they form consist of exactly two colours. Show that $n < 13$.

3. Let $\triangle ABC$ be a right-angled triangle with $\angle C = 90°$. CD is the altitude from C to AB, with D on AB. ω is the circumcircle of $\triangle BCD$. ω_1 is a circle situated in $\triangle ACD$, which is tangent to the segments AD and AC at M and N respectively, and is also tangent to circle ω.

   (i) Show that $BD \cdot CN + BC \cdot DM = CD \cdot BM$.

   (ii) Show that $BM = BC$.

4. Find all nonnegative integers m and n that satisfy the equation:

$$107^{56}(m^2 - 1) - 2m + 3 = \binom{113^{114}}{n}.$$

(If n and r are nonnegative integers satisfying $r \leqslant n$, then $\binom{n}{r} = C_r^n = \dfrac{n!}{r!(n-r)!}$ and $\binom{n}{r} = 0$ if $r > n$.)

Year 2010

1. Let $\triangle ABC$ be a scalene triangle. On the sides of $\triangle ABC$ three regular n-gons, external to the triangle, are drawn. Find all values of n for which the centres of the n-gons are the vertices of an equilateral triangle.

2. Let n be a positive integer. Determine the number of sequences $x_1, x_2, \ldots, x_{2n-1}, x_{2n}$, with $x_i = 1$ or -1, and such that $\left|\sum_{i=2k-1}^{2m} x_i\right| \leqslant 2$ for all integers k, m, with $1 \leqslant k \leqslant m \leqslant n$.

3. Let n be a positive integer and let a be an integer with $\gcd(a, n) = 1$. Prove that
$$\frac{a^{\varphi(n)} - 1}{n} \equiv \sum_{\substack{j=1 \\ \gcd(j,n)=1}}^{n} \frac{1}{aj}\left[\frac{aj}{n}\right] \pmod{n},$$
where $\varphi(n)$ is the Euler's totient function and $[x]$ is the greatest integer not exceeding x.

4. For any positive real numbers a, b, c, prove that
$$(ab + bc + ca)\left(\frac{1}{(a+b)^2} + \frac{1}{(b+c)^2} + \frac{1}{(c+a)^2}\right) \geqslant \frac{9}{4}.$$

Year 2011

1. Let x_1 be a positive real number and

$$x_{n+1} = \sqrt{5}x_n + 2\sqrt{x_n^2 + 1} \quad \text{for } n = 1, 2, 3, \ldots.$$

   Prove that among $x_1, x_2, \ldots, x_{2011}$, there are at least 670 irrational numbers.

2. The incircle of a triangle ABC touches the three sides BC, CA and AB at D, E and F respectively. The line segments BE and CF intersect the incircle at Y and Z respectively. Suppose the lines FE and BC intersect externally at X. Prove that the three points X, Y and Z are collinear.

3. A finite sequence of integers $a_0, a_1, \ldots, a_n$ is *quadratic* if for each $i = 1, 2, \ldots, n$, $|a_i - a_{i-1}| = i^2$.

   (a) Show that for any two integers b and c, $b < c$, there exists a natural number n and a quadratic sequence with $a_0 = b$ and $a_n = c$.

   (b) Find the smallest natural number n for which there exists a quadratic sequence with $a_0 = 0$ and $a_n = 2012$.

4. Let $A_1A_2 \cdots A_n$ be a cyclic polygon ($n \geqslant 3$). Find the maximum number of distinct acute-angled triangles whose vertices are chosen from $A_1, A_2, \ldots, A_n$.

Year 2012

1. For any positive integer n, let $a_1, a_2, \ldots, a_m$ be all the positive divisors of n, where $m \geqslant 1$. If there exist m integers $b_1, b_2, \ldots, b_m$ such that $n = \sum_{i=1}^{m} (-1)^{b_i} a_i$, then we say that n is a *good number*. Prove that there exists a good number with exactly 2013 distinct prime factors.

2. Some of the lattice points (x, y), with $1 \leqslant x \leqslant 101$ and $1 \leqslant y \leqslant 101$ are marked so that no 4 marked points form the vertices of an isosceles trapezoid with bases parallel to the x-axis or the y-axis (a rectangle is counted as an isosceles trapezoid). Determine the maximum number of marked points. (A lattice point is a point with integral coordinates.)

3. Prove that for every positive integer n and every group of real numbers $a_1, a_2, \ldots,$ $a_n > 0$,
$$\sum_{k=1}^{n} \frac{k}{a_1^{-1} + a_2^{-1} + \cdots + a_k^{-1}} \leqslant 2 \sum_{k=1}^{n} a_k.$$
Can "2" immediately to the right of the inequality be replaced by a smaller positive number?

4. In $\triangle ABC$, $AB > AC$, M is the midpoint of $\widehat{BC}$ of its circumcircle containing A. Its incircle with incentre I is tangent to BC at D. The line passing through D and parallel to AI intersects the incircle again at P. Prove that the lines AP and IM intersect at a point on the circumcircle of $\triangle ABC$.

Year 2013

1. Let a, b and c be positive real numbers such that $ab + bc + ca = 1$. Prove that

$$\sqrt[4]{\frac{\sqrt{3}}{a} + 6\sqrt{3}b} + \sqrt[4]{\frac{\sqrt{3}}{b} + 6\sqrt{3}c} + \sqrt[4]{\frac{\sqrt{3}}{c} + 6\sqrt{3}a} \leqslant \frac{1}{abc}.$$

   When does the equality hold?

2. For any positive integer a, define $M(a)$ to be the number of positive integers b for which $a + b$ divides ab. Find all integer(s) a with $1 \leqslant a \leqslant 2013$, so that $M(a)$ is largest possible in the range of a.

3. Given $\triangle ABC$ with $CA > BC > AB$, let O and H be the circumcentre and orthocentre of $\triangle ABC$ respectively. Denote by D and E the midpoints of arcs $\widehat{AB}$ and $\widehat{AC}$ of the circumcircle of $\triangle ABC$ not containing the opposite vertices. Let D' be the reflection of D in side AB and E' be the reflection of E in side AC. Prove that O, H, D', E' are concyclic if and only if A, D', E' are collinear.

4. In a chess tournament there are n players (where $n > 1$ is odd), and every two players play against each other exactly once. It is known that exactly n games end in a tie. For any set S of players including A and B, we say that A *admires* B in S if

   (a) A does not beat B; or

   (b) there exists a sequence of other distinct players $C_1, C_2, \ldots, C_k$ in S such that A does not beat C_1; C_k does not beat B; and C_i does not beat C_{i+1} for $1 \leqslant i \leqslant k - 1$.

   A set of four players is said to be *harmonic* if each of the four players admires everyone else in the set. Find (in terms of n) the greatest possible number of harmonic sets.

Year 2014

1. A polygon is *monochromatic* if all its vertices are coloured by a same colour. Suppose now every point of the plane is coloured red or blue. Show that there exists either a monochromatic equilateral triangle of side length 2, or a monochromatic equilateral triangle of side length $\sqrt{3}$, or a monochromatic rhombus of side length 1.

2. Let a, b, c be distinct nonzero real numbers. If the equations $ax^3 + bx + c = 0$, $bx^3 + cx + a = 0$ and $cx^3 + ax + b = 0$ have a common root, prove that at least one of these equations has three real roots (not necessarily distinct).

3. Find all pairs (a, b) of integers a and b satisfying

$$(b^2 + 11(a - b))^2 = a^3 b.$$

4. Let ABC be a scalene triangle, and let D and E be points on sides AB and AC respectively such that the circumcircles of triangles ACD and ABE are tangent to BC. Let F be the intersection point of BC and DE. Prove that AF is perpendicular to the Euler line of ABC.

   (Recall that the Euler line of a triangle is the line passing through its circumcentre and the orthocentre.)

Year 2015

1. Let $a_1, a_2, \ldots, a_n$ be a sequence of real numbers lying between 1 and -1, i.e. $-1 < a_i < 1$, for $1 \leqslant i \leqslant n$, and such that

   (i) $a_1 + a_2 + \cdots + a_n = 0$;

   (ii) $a_1^2 + a_2^2 + \cdots + a_n^2 = 40$.

   Determine the smallest possible value of n.

2. Find all integral ordered triples (x, y, z) such that $\sqrt{\dfrac{2015}{x+y}} + \sqrt{\dfrac{2015}{y+z}} + \sqrt{\dfrac{2015}{z+x}}$ are positive integers.

3. Let ABC be a triangle. Let D and E be respectively points on the segments AB and AC, and such that $DE // BC$. Let M be the midpoint of BC. Let P be a point such that $DB = DP$, $EC = EP$ and such that the open segments (segments excluding the endpoints) AP and BC intersect. Suppose $\angle BPD = \angle CME$. Show that $\angle CPE = \angle BMD$.

4. Let $n \geqslant 2$ be an integer. There are n distinct circles on the plane such that any two circles have two distinct intersections and no three circles have a common intersection. Initially there is a coin on each of the intersection points of the circles. Starting from X, players X and Y alternately take away a coin, with the restriction that one cannot take away a coin lying on the same circle as the last coin just taken away by the opponent in the previous step. The one who cannot do so will lose. In particular, one loses when there is no coin left. For what values of n does Y has a winning strategy?

Year 2016

1. A, B and C are three persons among a set P of n $(n \geqslant 3)$ persons. It is known that A, B and C are friends of one another, and that every one of the three persons has already made friends with more than half the total number of people in P. Given that every three persons who are friends of one another form a *friendly group*, what is the minimum number of friendly groups that may exist in P?

2. Let k be a positive integer. Find the number of nonnegative integers n less than or equal to 10^k satisfying the following conditions:

   (i) n is divisible by 3;

   (ii) Each decimal digit of n is one of the digits $2, 0, 1$ or 7.

3. Let $\triangle ABC$ be an acute-angled triangle. Let D be a point on the segment BC, I the incentre of $\triangle ABC$. The circumcircle of $\triangle ABD$ meets BI at P and the circumcircle of $\triangle ACD$ meets CI at Q. If the area of $\triangle PID$ and the area of $\triangle QID$ are equal, prove that $PI \times QD = QI \times PD$.

4. Let λ be a nonnegative real number and such that $\dfrac{a+b}{2} \geqslant \lambda\sqrt{ab} + (1-\lambda)\sqrt{\dfrac{a^2+b^2}{2}}$ holds for all positive real numbers a and b. Find the smallest possible value of λ.

Year 2017

1. The sequence $\{x_n\}$ is defined by $x_1 = 5$ and $x_{k+1} = x_k^2 - 3x_k + 3$ for $k = 1, 2, 3, \ldots$. Prove that $x_k > 3^{2^{k-1}}$ for any positive integer k.

2. Suppose $ABCD$ is a cyclic quadrilateral. Extend DA and DC to P and Q respectively such that $AP = BC$ and $CQ = AB$. Let M be the midpoint of PQ. Show that $MA \perp MC$.

3. Let k be a positive integer. Prove that there exists a positive integer ℓ with the following property: if m and n are positive integers relatively prime to ℓ such that $m^m \equiv n^n \pmod{\ell}$, then $m \equiv n \pmod{k}$.

4. Suppose 2017 points in a plane are given such that no three points are collinear. Among the triangles formed by any three of these 2017 points, those triangles having the largest area are said to be *good*. Prove that there cannot be more than 2017 good triangles.

Year 2018

1. Given that a, b and c are positive real numbers such that $ab + bc + ca \geqslant 1$, prove that
$$\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \geqslant \frac{\sqrt{3}}{abc}.$$

2. Find the number of nonnegative integers k, $0 \leqslant k \leqslant 2188$, and such that $\binom{2188}{k}$ is divisible by 2188.

   (Note that the binomial coefficient is defined by $\binom{n}{r} = \frac{n!}{r!(n-r)!}$.)

3. The incircle of $\triangle ABC$, with incentre I, meets BC, CA and AB at D, E and F respectively. The line EF cuts the lines BI, CI, BC and DI at points K, L, M and Q respectively. The line through the midpoint of CL and M meets CK at P.

   (a) Determine $\angle BKC$.

   (b) Show that the lines PQ and CL are parallel.

4. Find all integers $n \geqslant 3$ with the following property: there exist n distinct points on the plane such that each point is the circumcentre of a triangle formed by 3 of the points.

Year 2019

1. Given that $\{a_n\}$ and $\{b_n\}$ are two sequences of integers defined by

$$\begin{aligned} a_1 = 1,\ a_2 = 10,\ a_{n+1} = 2a_n + 3a_{n-1} \quad &\text{for } n = 2, 3, 4, \ldots, \\ b_1 = 1,\ b_2 = 8,\ b_{n+1} = 3b_n + 4b_{n-1} \quad &\text{for } n = 2, 3, 4, \ldots. \end{aligned}$$

   Prove that, besides the number '1', no two numbers in the sequences are identical.

2. Let $S = \{1, 2, \ldots, 100\}$. Consider a partition of S into $S_1, S_2, \ldots, S_n$ for some n, i.e. S_i are nonempty, pairwise disjoint and $S = \bigcup_{i=1}^{n} S_i$. Let a_i be the average of elements of the set S_i. Define the score of this partition by

$$\frac{a_1 + a_2 + \cdots + a_n}{n}.$$

   Among all n and partitions of S, determine the minimum possible score.

3. Let $\triangle ABC$ be an isosceles triangle with $AB = AC$. The incircle Γ of $\triangle ABC$ has centre I, and it is tangent to the sides AB and AC at F and E respectively. Let Ω be the circumcircle of $\triangle AFE$. The two external common tangents of Γ and Ω intersect at a point P. If one of these external common tangents is parallel to AC, prove that $\angle PBI = 90°$.

4. There are $n \geqslant 3$ cities in a country and between any two cities A and B, there is either a one way road from A to B, or a one way road from B to A (but never both). Assume the roads are built such that it is possible to get from any city to any other city through these roads, and define $d(A, B)$ to be the minimum number of roads you must go through to go from city A to B. Consider all possible ways to build the roads. Find the minimum possible average value of $d(A, B)$ over all possible ordered pairs of distinct cities in the country.

Year 2020

1. There is a table with n rows and 18 columns. Each of its cells contains a 0 or a 1. The table satisfies the following properties:

   (i) Every two rows are different.

   (ii) Each row contains exactly 6 cells that contain 1.

   (iii) For every three rows, there exists a column so that the intersection of the column with the three rows (the three cells) all contain 0.

   What is the greatest possible value of n?

2. For each positive integer n larger than 1 with prime factorization $p_1^{\alpha_1}p_2^{\alpha_2}\cdots p_k^{\alpha_k}$, its *signature* is defined as the sum $\alpha_1+\alpha_2+\cdots+\alpha_k$. Does there exist 2020 consecutive positive integers such that among them, there are exactly 1812 integers whose signatures are strictly smaller than 11?

3. Let $ABCD$ be a cyclic quadrilateral inscribed in a circle Γ such that $AB = AD$. Let E be a point on the segment CD such that $BC = DE$. The line AE intersects Γ again at F. The chords AC and BF meet at M. Let P be the symmetric point of C about M. Prove that PE and BF are parallel.

4. Let a, b and c be positive real numbers satisfying $abc = 1$. Prove that

$$\frac{1}{a^3+2b^2+2b+4}+\frac{1}{b^3+2c^2+2c+4}+\frac{1}{c^3+2a^2+2a+4}\leqslant\frac{1}{3}.$$

Year 2021

1. Is it possible to find a non-constant polynomial $P(x, y)$ such that $P(\lfloor\alpha\rfloor, \lfloor 3\alpha\rfloor) = 0$ for every real number α? (Here $\lfloor\mu\rfloor$ stands for the largest integer less than or equal to μ.)

2. In acute $\triangle ABC$, $BC > AB$. Let D be the point on the side AC such that $BD = BA$. Let M be the midpoint of BC. The tangents at D and M to the circumcircle of $\triangle CDM$ intersect at point E. The line BE intersects the side AC at F. Prove that A, B, M, F are concyclic.

3. Alice and Bob play a game on the plane. Firstly, Alice draws 4 red points which are the vertices of a square, and draws another 2021 red points inside this square such that no three red points are collinear. Next, Bob draws n blue points such that for every triangle whose three vertices are the red points, there is at least one blue point lying strictly inside this triangle.

   The goal of Alice is to maximize n, while the goal of Bob is to minimize n. Find the value of n if both players play optimally.

4. Let $f(n) = \prod_{k=1}^{n}\left(1 + 4\cos^2\left(\frac{k\pi}{2n+1}\right)\right)$. Prove that $f(n)$ is an integer for all positive integers n.

Year 2022

1. Suppose a, b and c are nonzero real numbers satisfying $abc = 2$. Prove that among the three numbers $2a - \dfrac{1}{b}$, $2b - \dfrac{1}{c}$ and $2c - \dfrac{1}{a}$, at most two of them are greater than 2.

2. Find the period of the repetend of the fraction $\dfrac{39}{1428}$ using *binary* numbers, i.e. its binary decimal representation.

   (Note: When a proper fraction is expressed as a decimal number (of any base), either the decimal number terminates after finite steps, or it is of the form

   $$0.b_1b_2\cdots b_s a_1a_2\cdots a_k a_1a_2\cdots a_k a_1a_2\cdots a_k\cdots.$$

   Here the repeated sequence $a_1a_2\cdots a_k$ is called the *repetend* of the fraction, and the smallest length of the repetend, k, is called the *period* of the decimal number.)

3. Given a 2023×2023 square grid, there are beetles on some of the unit squares, with at most one beetle on each unit square. In the first minute, every beetle will move one step to its right or left adjacent square, or to its top or bottom adjacent square. In the second minute, every beetle will move again, only this time, in case the beetle moved right or left in the previous minute, it moves to top or bottom in this minute, and vice versa, and so on. What is the minimum number of beetles on the square grid to ensure that, no matter where the beetles are initially, and how they move in the first minute, but after finitely many minutes, at least two beetles will meet at a certain unit square?

4. Let $ABCD$ be a quadrilateral inscribed in a circle Γ such that $AB = BC = CD$. Let M and N be the midpoints of AD and AB respectively. The line CM meets Γ again at E. Prove that the tangent at E to Γ, the line AD and the line CN are concurrent.